

Derivation of Effective Pitch/Roll Stiffness and Damping for the Reduced Vehicle Model with Front/Rear Anti-Roll Bars

Prepared for the current MATLAB/CasADi reduced vehicle model

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Purpose

This note derives the effective parameters used in the reduced model:

$$K_\theta, \quad C_\theta, \quad K_\phi, \quad C_\phi,$$

where C_θ is the coefficient multiplying $\dot{\theta}$, and C_ϕ is the coefficient multiplying $\dot{\phi}$. The derivation also shows how front and rear anti-roll-bar (ARB) stiffnesses from CarSim should be added when CarSim defines them as axle roll stiffnesses in N m/rad.

The result is intended to be compatible with the current reduced MATLAB/CasADi model that has one pitch DOF and one roll DOF for the sprung mass, but no explicit four suspension-travel states.

1 Modelling assumptions and sign conventions

1.1 Reduced coordinates

The reduced model uses:

$$\theta = \text{sprung-mass pitch angle}, \quad \phi = \text{sprung-mass roll angle}.$$

The current sign convention is taken to be:

- positive θ : front suspension compression and rear suspension extension;
- positive ϕ : right side down / right side load increases;
- positive lateral acceleration a_y : produces positive roll response in the model;
- positive roll load-transfer moment increases right wheel load and decreases left wheel load.

The exact visual sign convention can be changed, but the stiffness and damping magnitudes derived below remain the same as long as the same convention is used in both the roll ODE and the wheel-load equations.

1.2 Corner vertical rates

Let the CarSim-equivalent wheel-centre vertical rates be

$$k_{w,FL}, k_{w,FR}, k_{w,RL}, k_{w,RR}$$

and the wheel-centre vertical damping rates be

$$c_{w,FL}, c_{w,FR}, c_{w,RL}, c_{w,RR}.$$

These must be wheel-centre rates. If CarSim gives spring/damper element rates instead, first convert by

$$k_{w,i} = k_{s,i} M R_i^2, \quad c_{w,i} = c_{s,i} M R_i^2,$$

where

$$M R_i = \frac{\text{spring or damper element displacement}}{\text{wheel-centre vertical displacement}}.$$

1.3 Geometry

Use the same symbols as the MATLAB code:

$$\begin{aligned} a &= \text{veh.acom}, & b &= \text{veh.bcom}, \\ t_F &= \text{veh.tF}, & t_R &= \text{veh.tR}. \end{aligned}$$

For the four corners, define longitudinal and lateral lever arms:

corner	FL	FR	RL	RR
x_i	a	a	$-b$	$-b$
y_i	$-t_F/2$	$+t_F/2$	$-t_R/2$	$+t_R/2$

With small pitch and roll angles, the vertical suspension deflection at each wheel centre is approximated by

$$z_i = x_i\theta + y_i\phi.$$

This is the linear small-angle approximation consistent with the reduced model.

2 General stiffness derivation from potential energy

The suspension spring potential energy is

$$U_{spr} = \frac{1}{2} \sum_i k_{w,i} z_i^2 = \frac{1}{2} \sum_i k_{w,i} (x_i\theta + y_i\phi)^2.$$

Expanding gives

$$U_{spr} = \frac{1}{2} K_{\theta,spr} \theta^2 + K_{\theta\phi,spr} \theta\phi + \frac{1}{2} K_{\phi,spr} \phi^2,$$

where

$$\begin{aligned} K_{\theta,spr} &= \sum_i k_{w,i} x_i^2, \\ K_{\phi,spr} &= \sum_i k_{w,i} y_i^2, \\ K_{\theta\phi,spr} &= \sum_i k_{w,i} x_i y_i. \end{aligned}$$

The current reduced model keeps only diagonal pitch and roll restoring terms, not a pitch-roll coupling term. If the left and right rates are symmetric on each axle,

$$k_{w,FL} = k_{w,FR}, \quad k_{w,RL} = k_{w,RR},$$

then $K_{\theta\phi,spr} = 0$. If the vehicle is left/right asymmetric, ignoring $K_{\theta\phi,spr}$ is a modelling approximation.

3 Effective pitch stiffness

Using $x_{FL} = x_{FR} = a$ and $x_{RL} = x_{RR} = -b$,

$$K_\theta = K_{\theta,spr} = (k_{w,FL} + k_{w,FR})a^2 + (k_{w,RL} + k_{w,RR})b^2.$$

This is the equation implemented in the parameter helper:

```
veh.Ktheta = (kwFL + kwFR)*a^2 + (kwRL + kwRR)*b^2;
```

The pitch restoring moment used by the model is

$$M_{\theta,rest} = K_\theta\theta + C_\theta\dot{\theta},$$

and the pitch ODE applies it with a negative sign:

$$I_y\ddot{\theta} = -m_s h_s a_x + M_{\theta,ext} - C_\theta\dot{\theta} - K_\theta\theta.$$

4 Effective pitch damping

Use the Rayleigh dissipation function:

$$D = \frac{1}{2} \sum_i c_{w,i} \dot{z}_i^2.$$

For pitch-only motion, $\dot{z}_{FL} = a\dot{\theta}$, $\dot{z}_{FR} = a\dot{\theta}$, $\dot{z}_{RL} = -b\dot{\theta}$, and $\dot{z}_{RR} = -b\dot{\theta}$. Therefore

$$C_\theta = (c_{w,FL} + c_{w,FR})a^2 + (c_{w,RL} + c_{w,RR})b^2.$$

This is implemented as

```
veh.Ctheta = (cwFL + cwFR)*a^2 + (cwRL + cwRR)*b^2;
```

5 Effective roll stiffness from corner suspension rates

For pure roll,

$$\begin{aligned} z_{FL} &= -\frac{t_F}{2}\phi, & z_{FR} &= +\frac{t_F}{2}\phi, \\ z_{RL} &= -\frac{t_R}{2}\phi, & z_{RR} &= +\frac{t_R}{2}\phi. \end{aligned}$$

Thus the front and rear suspension roll stiffness contributions are

$$\begin{aligned} K_{\phi,F,spr} &= (k_{w,FL} + k_{w,FR}) \left(\frac{t_F}{2} \right)^2, \\ K_{\phi,R,spr} &= (k_{w,RL} + k_{w,RR}) \left(\frac{t_R}{2} \right)^2. \end{aligned}$$

The spring-only total roll stiffness is

$$K_{\phi,spr} = K_{\phi,F,spr} + K_{\phi,R,spr}.$$

These equations are implemented as

```
veh.Kphi_F_spr = (kwFL + kwFR)*(tF/2)^2;
veh.Kphi_R_spr = (kwRL + kwRR)*(tR/2)^2;
```

The grouping is by axle, $(FL + FR)$ and $(RL + RR)$, because roll is a left-right motion within each axle. The grouping $(FL + RL)$ and $(FR + RR)$ is a side grouping and is not the correct decomposition for front/rear roll-stiffness distribution.

6 Effective roll damping from corner dampers

Using Rayleigh dissipation with pure roll velocities,

$$\begin{aligned} C_{\phi,F} &= (c_{w,FL} + c_{w,FR}) \left(\frac{t_F}{2} \right)^2, \\ C_{\phi,R} &= (c_{w,RL} + c_{w,RR}) \left(\frac{t_R}{2} \right)^2. \end{aligned}$$

Therefore

$$C_\phi = C_{\phi,F} + C_{\phi,R}.$$

This is implemented as

```
veh.Cphi_F = (cwFL + cwFR)*(tF/2)^2;
veh.Cphi_R = (cwRL + cwRR)*(tR/2)^2;
veh.Cphi   = veh.Cphi_F + veh.Cphi_R;
```

7 Adding front and rear anti-roll bars

Assume CarSim gives front and rear ARB stiffnesses directly as axle roll stiffnesses:

$$K_{ARB,F}, \quad K_{ARB,R} \quad [\text{N m/rad}].$$

Then the ARB strain energy is

$$U_{ARB} = \frac{1}{2}K_{ARB,F}\phi^2 + \frac{1}{2}K_{ARB,R}\phi^2.$$

Because the units are already N m/rad, no additional track-width conversion is applied. The front and rear total roll stiffnesses are

$$K_{\phi,F} = K_{\phi,F,spr} + K_{ARB,F},$$

$$K_{\phi,R} = K_{\phi,R,spr} + K_{ARB,R},$$

and the total roll stiffness is

$$K_{\phi} = K_{\phi,F} + K_{\phi,R}.$$

This is implemented as

```
veh.Kphi_F_arb    = veh.Karb_F;
veh.Kphi_R_arb    = veh.Karb_R;
veh.Kphi_F_total  = veh.Kphi_F_spr + veh.Kphi_F_arb;
veh.Kphi_R_total  = veh.Kphi_R_spr + veh.Kphi_R_arb;
veh.Kphi          = veh.Kphi_F_total + veh.Kphi_R_total;
```

The ARB is not added to pitch stiffness. In a pure pitch motion, the left and right wheels on the same axle move in the same direction, so an ideal anti-roll bar has no twist and no pitch restoring moment.

8 Roll ODE consistency

The current model obtains the front and rear roll stiffnesses and dampings as

```
Kphi_f = getvehfield_any(veh, {'Kphi_F_total', 'Kphi_f', 'Kphi_front'}, ...
    veh.Kphi * veh.dis_Kphi_ratio);
Kphi_r = getvehfield_any(veh, {'Kphi_R_total', 'Kphi_r', 'Kphi_rear'}, ...
    veh.Kphi * (1 - veh.dis_Kphi_ratio));

Cphi_f = getvehfield_any(veh, {'Cphi_F', 'Cphi_f', 'Cphi_front'}, ...
    veh.Cphi * veh.dis_Cphi_dot_ratio);
Cphi_r = getvehfield_any(veh, {'Cphi_R', 'Cphi_r', 'Cphi_rear'}, ...
    veh.Cphi * (1 - veh.dis_Cphi_dot_ratio));

Kphi_total = Kphi_f + Kphi_r;
Cphi_total = Cphi_f + Cphi_r;
```

The roll equation is then

$$I_x \ddot{\phi} = m_s h_{roll} a_y - C_{\phi,total} \dot{\phi} - K_{\phi,total} \phi,$$

where

$$K_{\phi,total} = K_{\phi,F} + K_{\phi,R}, \quad C_{\phi,total} = C_{\phi,F} + C_{\phi,R}.$$

This guarantees that the single roll DOF sees the same total restoring moment as the wheel-load calculation.

9 Wheel-load transfer consistency

The elastic roll moments passed to the wheel-load function are

$$M_{el,F} = K_{\phi,F}\phi + C_{\phi,F}\dot{\phi},$$

$$M_{el,R} = K_{\phi,R}\phi + C_{\phi,R}\dot{\phi}.$$

In code:

```
M_el_F = Kphi_f * phi_roll + Cphi_f * phi_roll_dot;
M_el_R = Kphi_r * phi_roll + Cphi_r * phi_roll_dot;
```

For each axle,

$$M_{LT,axle} = \frac{t_{axle}}{2}(F_{z,R} - F_{z,L}).$$

Therefore the per-wheel load increments are

$$\Delta F_{z,R} = +\frac{M_{LT,axle}}{t_{axle}}, \quad \Delta F_{z,L} = -\frac{M_{LT,axle}}{t_{axle}}.$$

This matches the implemented equations:

```
Fz_FL = 0.5 * Fz_front_total - M_LT_F / tF;
Fz_FR = 0.5 * Fz_front_total + M_LT_F / tF;
Fz_RL = 0.5 * Fz_rear_total - M_LT_R / tR;
Fz_RR = 0.5 * Fz_rear_total + M_LT_R / tR;
```

Thus a positive front roll-transfer moment increases front-right load and decreases front-left load. The same convention is used at the rear axle.

10 Complete parameter mapping used by the code

The parameter helper in `my_params.m` implements this mapping:

$$\begin{aligned} K_{\theta} &= (k_{w,FL} + k_{w,FR})a^2 + (k_{w,RL} + k_{w,RR})b^2, \\ C_{\theta} &= (c_{w,FL} + c_{w,FR})a^2 + (c_{w,RL} + c_{w,RR})b^2, \\ K_{\phi,F,spr} &= (k_{w,FL} + k_{w,FR})(t_F/2)^2, \\ K_{\phi,R,spr} &= (k_{w,RL} + k_{w,RR})(t_R/2)^2, \\ C_{\phi,F} &= (c_{w,FL} + c_{w,FR})(t_F/2)^2, \\ C_{\phi,R} &= (c_{w,RL} + c_{w,RR})(t_R/2)^2, \\ K_{\phi,F} &= K_{\phi,F,spr} + K_{ARB,F}, \\ K_{\phi,R} &= K_{\phi,R,spr} + K_{ARB,R}, \\ K_{\phi} &= K_{\phi,F} + K_{\phi,R}, \\ C_{\phi} &= C_{\phi,F} + C_{\phi,R}, \\ \text{dis_Kphi_ratio} &= \frac{K_{\phi,F}}{K_{\phi}}, \\ \text{dis_Cphi_dot_ratio} &= \frac{C_{\phi,F}}{C_{\phi}}. \end{aligned}$$

11 Important limitations

The derivation is correct for the reduced model under the stated assumptions, but it is not a full CarSim suspension model. The following effects are not represented explicitly:

- nonlinear bump/rebound damping and damper velocity maps;
- bump stops, rebound stops, and nonlinear spring curves;
- roll-centre migration and suspension kinematic/compliance changes;
- anti-dive and anti-squat geometry;
- pitch-roll coupling from left/right asymmetric suspension rates;
- heave DOF and four individual suspension travel states;
- nonlinear ARB link geometry.

For black-box CarSim matching, the equations above should be treated as a physically correct initial estimate. Fine tuning against CarSim manoeuvres may still be needed.

12 Recommended sanity checks

Before fitting against CarSim, check the following:

1. If all four ARB values are set to zero, the roll stiffness should reduce to the suspension spring roll stiffness only.
2. Changing $K_{ARB,F}$ should change front elastic lateral load transfer but not pitch stiffness.
3. Changing $K_{ARB,R}$ should change rear elastic lateral load transfer but not pitch stiffness.
4. If $k_{w,FL} = k_{w,FR}$ and $k_{w,RL} = k_{w,RR}$, the pitch-roll cross stiffness is zero.
5. The roll ODE must use the same $K_{\phi,F} + K_{\phi,R}$ and $C_{\phi,F} + C_{\phi,R}$ that are used in the wheel-load transfer equations.

Conclusion

With wheel-centre vertical corner rates and CarSim ARB roll stiffnesses in N m/rad, the implemented equations are physically and mathematically consistent with the reduced model. The anti-roll bars should be added only to the front/rear roll stiffnesses, not to pitch stiffness, and the resulting front/rear roll contributions should be used both in the roll ODE and in the elastic lateral load-transfer equations.